

Supervisor Meeting Agenda

30 March 2026 — 30 minutes

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1. Progress Update (5 min)

- **DONE:** Training pipeline verified end-to-end: 10 000-step smoke run completed successfully. Evidential metrics confirmed live during training (epistemic uncertainty = 1.44, aleatoric = 2.07). All three containers healthy throughout.
- **DONE:** Sensor-to-covariance pipeline fully validated. Dry-run inspection (8 episodes, 300 steps each) confirmed: EKF pose tracks CARLA ground truth to within ~ 1 m relative error; odom-frame target bay transform is geometrically correct (round-trip reconstruction error = 0.000 m); covariance inflation events (staleness, Cartographer jump) visible in log output.
- **DONE:** Critical pipeline bugs resolved since last meeting:
 - `imu0.config` yaw absolute flag (index 5) was incorrectly set to `true` — EKF was treating IMU as an absolute heading source. Fixed to `false`; only yaw rate (index 11) enabled.
 - `twist_in_odom_frame` parameter was defined in `ros2.config.yaml` but not wired into the launch file. Now propagated correctly to `CovarianceExtractorNode`.
 - Odom-frame target bay transform implemented (`_compute_target_bay_odom`): both vehicle pose and target are expressed in the same accumulated Cartographer odom frame, so mid-episode drift cancels in the relative target vector (`obs[15–17]`).
- **DONE:** Placement work synthesised. The placement objective (preliminary design of an uncertainty-aware RL policy) has been exceeded — a full working implementation was produced. Placement demo and one-page report are being prepared (see Section 2.1).
- **TODO:** Full 1M-step training run not yet launched — pipeline is verified and ready. Covariance variation under NPC vehicles and weather randomisation still to be confirmed during live training.
- **TODO:** Mapping runs for `trapezoid` and `irregular_a` floor plans still outstanding (~ 90 s each, `make docker-map LAYOUT=<name>`).

2. Discussion Points (15 min)

2.1 Placement Deliverables — Demo and One-Page Report

The placement brief (Lemonworx, 8 weeks) required a preliminary design for an uncertainty-aware RL policy. The implementation is substantially complete. Two placement deliverables remain:

Demo plan (approx. 10 minutes, live):

1. `make docker-up` — show the three-container stack is healthy.
2. `make docker-inspect INSPECT_LAYOUT=trapezoid` — CARLA window with parking lot geometry, patrol NPC, and pedestrians.

3. `make docker-inspect-sensors INSPECT_SUITE=suite_a` — LiDAR arc and IMU position on the ego vehicle, connecting sensor to uncertainty signal.
4. Code walkthrough: `evidential_policy.py` (NIG output head, epistemic/aleatoric decomposition) and `sb3_integration.py` (evidential regularisation term in PPO loss).
5. Live ROS 2 topic echo of `/ekf_uncertainty/covariance` showing uncertainty values changing as simulation conditions change.

One-page report structure: (1) literature gap identified; (2) preliminary architecture design produced (actor-only EDL + EKF covariance as explicit state input); (3) implementation delivered beyond the placement scope; (4) relevance to teleoperation cognitive load problem (uncertainty thresholds trigger operator handoff).

Q: Is this demo scope appropriate for the placement assessment, or should it focus more on the theoretical design decisions rather than the live system?

Q: The one-page report will reference the 2×2 ablation as the evaluation strategy. Should it also include a brief statement on the sim-to-real transfer plan (same Docker stack, physical BMW i3 sensors), or is that scope beyond the placement brief?

2.2 Covariance Variation — Sufficiency for Training

Dry-run inspection with random actions in a clear empty lot shows baseline std values of ~ 0.10 – 0.12 m (flat, as expected with no occlusion). Meaningful covariance variation requires NPC vehicles and weather randomisation active during a real training run. This has not yet been observed live.

Q: Is it acceptable to begin the full 1M-step training run before separately verifying covariance variation, or should a short dedicated verification run (NPC vehicles + HardRainNoon, monitor covariance only, no training) be conducted first to confirm the signal is present?

Q: If covariance variation proves insufficient under the current sensor suite (Suite A: 2D LiDAR only), what is the preferred fallback: (a) Suite B (16-channel 3D LiDAR, wider scan coverage), (b) artificial noise injection on the EKF input, or (c) additional NPC density beyond the current maximum of 3 patrol vehicles and 4 pedestrians?

2.3 Cartographer Odom Frame Accumulation Across Episodes

The Cartographer odom frame accumulates continuously across training episodes (it is a process-level frame, not per-episode). As a result, absolute odom coordinates grow over many episodes. This is not a bug — the relative target vector (`obs[15–17]`) remains correct because drift cancels — but it means the raw pose values in `obs[0–2]` grow without bound.

`VecNormalize` handles this via running statistics. The policy never sees unbounded values.

Q: Is it acceptable to let absolute odom coordinates grow unbounded (mitigated by `VecNormalize`), or is a per-episode odom frame reset needed? Resetting would require restarting the Cartographer ROS process per episode (~ 10 – 30 s overhead), which is likely impractical for 1M-step training.

3. Timeline and Next Steps (5 min)

Q: Does this timeline remain on track for the September 2026 submission? Are there any tasks that should be deprioritised or cut if the ablation study takes longer than expected?

Task	Est. Time	Status
Mapping runs (trapezoid, irregular.a)	0.5 days	Ready to run
Full 1M-step training run	2–4 days (GPU)	Unblocked
Covariance variation verification	During training	Passive monitoring
Reward tuning (Task 9)	1–2 weeks	After first run
Full ablation (4 baselines $\times$ 10 seeds)	2–4 weeks	After Task 9
Tasks 10–14: Evaluation + plots	2 weeks	After ablation
Placement demo + one-page report	2–3 days	In progress
Dissertation writing	Ongoing	Can start now

4. Dissertation Writing (5 min)

- The methodology chapter is ready to draft: architecture is stable, all design decisions are resolved, and the sensor-to-covariance pipeline is fully documented.
- The background chapter (EKF, evidential deep learning, PPO, Cartographer) is independent of experimental results and can be written in parallel with training.
- The experimental setup chapter is also ready: floor plans, sensor suite, NPC configuration, and the Cartographer pure localisation workflow are all fixed.

Q: Any guidance on chapter structure, expected length, and when to submit draft chapters for feedback? Should chapters be submitted individually as they are written, or as a complete draft?

Q: The placement proposal and dissertation proposal share the same research problem and evaluation strategy. Should the dissertation introduction explicitly acknowledge the placement as the origin of the implementation, or treat the two submissions as entirely separate documents?